Shayan Shakeri

+1-416-471-6651 | shayansnezhad@gmail.com | github.com/sshakerinezhad | sshakerinezhad.github.io | merlyn-labs.com |
Toronto, Canada

PROFESSIONAL SUMMARY

AI Research Engineer focused on LLM and agent evaluation: designing hard-to-game LLM/VLM judges and deterministic eval pipelines that surface where frontier models fail. At BMO's AI Centre of Excellence, built a deterministic agent eval pipeline that uncovered systematic risk-downplaying bias in a GenAI tool serving $200B+ AUM; co-founded Merlyn Labs, developing VLM judges that turn model rollouts into dense rewards engineered to resist reward gaming (8th place, Stanford BEHAVIOR-1K Challenge).

CORE COMPETENCIES

LLM & Model Evaluation | LLM & VLM Judges | Benchmark Design | Evaluation Datasets & Environments | Reward-Gaming Resistance | Evaluation Infrastructure | Agent Evaluation | Alignment Evaluation

WORK EXPERIENCE

BMO AI Centre of Excellence
Sep 2024 - Present

AI Research Engineer Toronto

- Built a **deterministic agent evaluation pipeline** from hundreds of synthesized inputs to detect misaligned model outputs at scale.
- Uncovered systematic bias to downplay investment risk in a GenAI tool serving **$200B+ AUM** wealth management, using scalable eval tooling and synthesized evaluation datasets.
- Building a graph-based agentic system answering multi-hop relational queries across bank client data (agent evaluation over query paths).

Merlyn Labs
Aug 2025 - Present

AI Researcher & Co-Founder Toronto

- Developing **LLM/VLM judges** that score model rollouts into dense, context-dependent rewards designed to **resist reward gaming**.
- Showed pi0.5 LIBERO-PRO "generalization failures" are recipe-induced overfitting; proposed an alternative baseline.
- Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); placed 8th in Stanford's BEHAVIOR-1K Challenge with a published technical report.

Epineuron Technologies
May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op Mississauga

- Designed and assembled PCBs and authored validation protocols (IEC 60601-1, ISO 13485) for PeriPulse, an FDA Breakthrough-designated medical device now in multi-national clinical trials.

PROJECTS

Stanford BEHAVIOR-1K Challenge
8th Place, Standard Track

Benchmark failure-mode discovery and dataset construction: identified proprioceptive collapse (60% masking improved task success up to 48%) and showed chunked execution beat temporal ensembling 3x. Trained on 10,000+ demonstrations across 22 tasks; published a technical report and LessWrong analysis.

Evaluation methods, Failure-mode discovery, Dataset construction, Benchmark critique

EDUCATION

M.Eng, MIE - AI & Robotics, University of Toronto — Sep 2024 - Apr 2027

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op, McMaster University — Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

ML & AI: LLM Evaluation, LLM/VLM Judges, Benchmark Design, Evaluation Datasets & Environments, Alignment Evaluation, Agents, LLM Fine-tuning, RL, Imitation Learning, Python, C/C++, PyTorch, JAX

Robotics & Simulation: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2